

# Experimentally validated multiphysics modeling of fracture induced by thermal shocks in sintered $\text{UO}_2$ pellets

Levi D. McClenny<sup>a,1</sup>, Moiz I. Butt<sup>b,1</sup>, M. Gomaa Abdoelatef<sup>b,d,1</sup>, Michal J. Pate<sup>b</sup>, Kay L. Yee<sup>b</sup>, R. Harikrishnan<sup>c</sup>, Delia Perez-Nunez<sup>b</sup>, W. Jiang<sup>e</sup>, Luis H. Ortega<sup>b</sup>, Sean M. McDeavitt<sup>b</sup>, Karim Ahmed<sup>b,d,\*</sup>

<sup>a</sup> Department of Electrical Engineering, Texas A&M University, College Station, TX 77843, USA

<sup>b</sup> Department of Nuclear Engineering, Texas A&M University, College Station, TX 77843, USA

<sup>c</sup> Department of Aerospace Engineering, Texas A&M University, College Station, TX 77843, USA

<sup>d</sup> Department of Material Science and Engineering, Texas A&M University, College Station, TX 77843, USA

<sup>e</sup> Computational Mechanics and Materials, Idaho National Laboratory, P.O. Box 1625, Idaho Falls, ID 83415-3840, USA

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## ABSTRACT

Uranium Dioxide ( $\text{UO}_2$ ) fuel powers almost all commercial Nuclear Power Plants (NPPs) worldwide, generating carbon-free energy and contributing to the fight against climate change.  $\text{UO}_2$  fuel incurs damage and fractures due to large thermal gradients that develop across the fuel pellet during normal and transient operating conditions. A comprehensive understanding of the underlying mechanisms by which these processes take place is still lacking. A combined experimental and computational approach is utilized here to quantify the behavior of  $\text{UO}_2$  fuel fracture induced by thermal shock. This work introduces both (1) an experimental study to understand the fuel fracturing behavior of sintered  $\text{UO}_2$  pellets when exposed to thermal shock, and (2) a Multiphysics phase-field fracture model capable of simulating this process. Parametric studies were conducted to evaluate the effects of uncertainties in fracture properties on the fracture behavior of  $\text{UO}_2$  due to thermal shocking. A set of energy release rate (or equivalently fracture toughness) and contract area (the part of the fuel pellet in direct contact with the cold bath) were able to capture the overall fracture trends of the corresponding experimental data. Our combined approach presents a new method for accounting for the effects of microstructure and sample size on the energy release rate/fracture toughness. The experimental data were collected from multiple experiments that exposed  $\text{UO}_2$  pellets to high-temperature conditions (589–676 °C) followed by a quench in sub-zero water. This work demonstrates that joint experimental and computational efforts are able to advance the understanding of thermal fracture in the primary fuel source for existing and future NPPs.

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## 1. Introduction

Most commercial nuclear power plants in U.S. and worldwide use  $\text{UO}_2$  as fuel. Despite downsides with high-thermal strains due to poor thermal conductivity,  $\text{UO}_2$  is still the primary choice for LWRs due to its high melting point (2865 °C), lack of phase change up to its melting point (unlike metallic fuels), good corrosion resistance, and fission product retention [1]. Fuel fracturing impacts the thermo-mechanical properties of  $\text{UO}_2$  during normal and transient operation. Since  $\text{UO}_2$  has a very low thermal conductivity [2], a

large thermal gradient can build up (Soret effect) with a mass gradient leading to the initiation of pores.

In as-fabricated unirradiated LWR fuel, volumetric heating from fission is mostly spatially uniform (except on the pellet periphery). As burnup increases, self-shielding effects result in higher volumetric heat generation around the perimeter of the fuel pellets [3,4]. The heat is transferred radially outward from the fuel pellet, across the fuel-cladding gap, through the cladding, and finally into the coolant where it converts aqueous water into gaseous steam. During reactor operation, nuclear fuels experience swelling and form voids and bubbles, which may cause the pellet to fracture at higher temperatures. Fuel fracturing may occur when the fuel is thermally shocked, or a rapid decrease in temperature as a result of the thermo-mechanical stress incurred by the grain boundaries of the material and varies depending on density, poros-

\* Corresponding author at: Department of Nuclear Engineering, Texas A&M University, College Station, TX 77843, USA.

E-mail address: [karim.ahmed@tamu.edu](mailto:karim.ahmed@tamu.edu) (K. Ahmed).

<sup>1</sup> These authors contributed equally to this study.

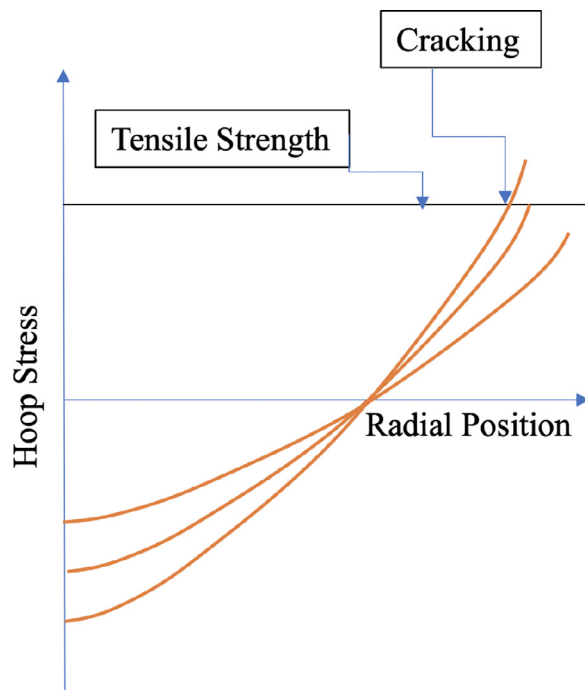


Fig. 1. Plot of variation in hoop stress in fresh LWR fuel as a function of radial position during ramping to full power [5].

ity, chemical composition, and microstructural state. Non-uniform thermal expansion in the fuel pellet causes compressive stresses in the center and tensile stresses on the exterior. This hoop stress as a function of radial position is pictured in Fig. 1. It can be observed from the plot that radial cracking occurs as the tensile hoop stress surpasses the tensile strength of the fuel due to increasing power [5]. The tensile strength of  $\text{UO}_2$  is about 150 MPa and fracture initiation is expected to occur when the hoop stress surpasses this value [5]. In this work, a coupled experimental and computational approach was utilized to investigate thermal fracture in  $\text{UO}_2$ . First, to induce fracture in  $\text{UO}_2$  pellets, the pellets were submerged in molten salts and heated to a temperature in the range of (589–676 °C). Then these pellets were subjected to a rapid temperature decrease to mimic accidents-like transients. Second, the computational model was used to simulate, reproduce, and interpret the experimental data.

## 2. Background

Fuel fracturing in LWRs is a well expected occurrence during reactor operation, however, limited experimental data with varying configurations (such as high-burnup fuel) of fracture initiation and propagation of  $\text{UO}_2$  fuel pellets exists. Since replication of a LWR environment for a fuel pellet is difficult due to the temperature profile being a result of a combination of volumetric fission heating and coolant heat removal, few have taken on this challenge [5]. With this limited data availability, novel modeling approaches cannot be implemented with certainty. This work results would add upon current literature data that allow further investigations for radial relocation, pellet-cladding mechanical interactions, and fragmentation phenomena to benefit fuel performance objectives in LWRs. To fulfill this requirement for further experimental validation, fuel fracturing experiments were carried out that subject the fuel to a temperature profile that allows for an observation of the cracking propagation and emphasizes fuel cracking rather than any unrelated phenomena.

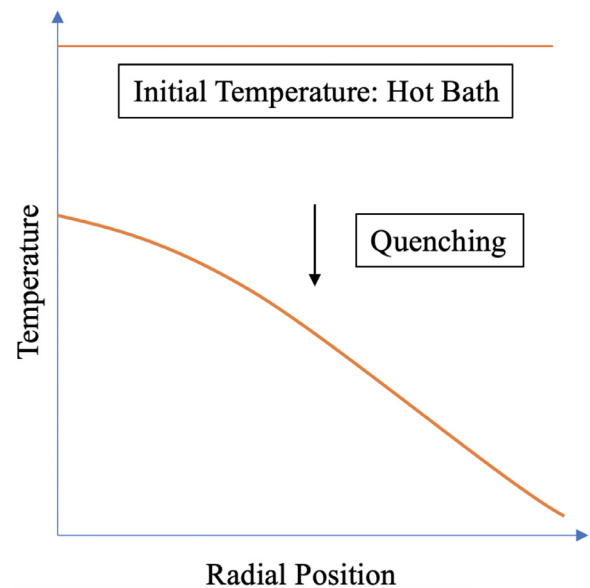


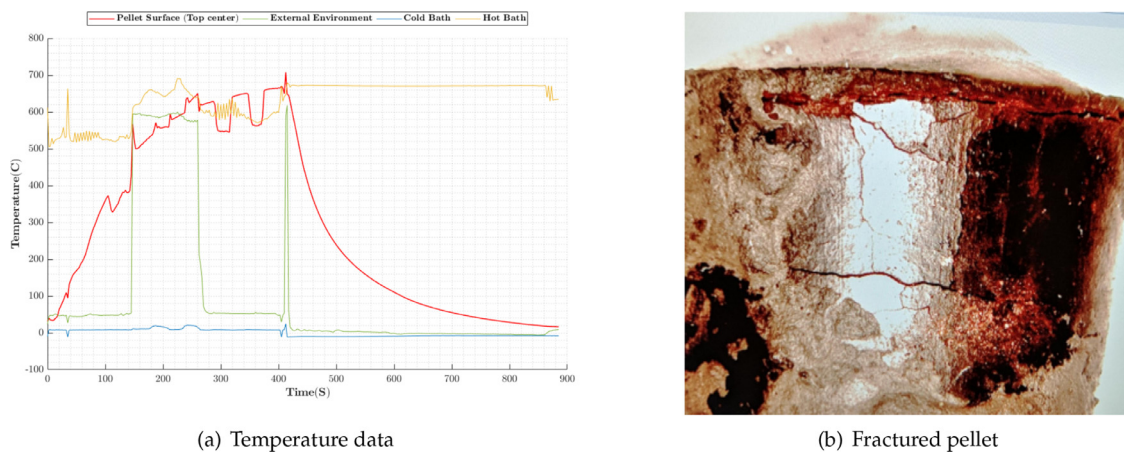
Fig. 2. Plot of expected temperature profile during quenching after achieving a spatially uniform high temperature in the pellet [5].

As part of a three-component experimental study focused on different heating methods of  $\text{UO}_2$  fuel, the purpose of this section was to slowly heat a fuel pellet contained in a metal tube assembly to a high temperature and subsequently quench said tube in a cold bath designed to insulate the top and bottom portions of the cylindrical pellet to generate a radial heat flux [5]. Fig. 2 displays an expected temperature profile of the quenching process when cooling from a spatially uniform high temperature. The  $\text{UO}_2$  pellets used for this study were manufactured at Texas A&M University's Fuel Cycle and Materials Laboratory (FCML) using powder metallurgical methods and sintering to produce cylindrical  $\text{UO}_2$  fuel pellets of approximately 9–10 mm in height and 10–11 mm diameter each. Fuel pellets of varying density, diameter, and heights were used in the thermal shocking experiments.

This contribution layout describes this work experiment method and procedure of the thermal-shock induced fracture in the  $\text{UO}_2$  in Section 3. This work model is explained in Section 4, in which the model parameters are calculated based on the  $\text{UO}_2$  thermal and mechanical properties available in the literature. Both experimental and modeling results were presented and discussed in Sections 5 and 6, respectively. Finally, Section 7 summarizes the research conducted in this study and highlights the possible ways for further studies directions.

## 3. Experimental methods

Experimental data has been collected on fuel fracturing of sintered  $\text{UO}_2$  pellets by thermally shocking them in high-temperature conditions ( $\sim 700$  °C ternary carbonate salt) and subsequently in sub-zero conditions ( $-10$  °C in water or 1:1 ethylene glycol:water). More detail about the experimental procedures can be found in Ortega et al. [6]. Here, we present multiphysics computational modeling approaches and additional experimental results, extending the work performed in Ortega et al. [6] and adding supplemental discussion. The experimental set-up consists of a  $\text{UO}_2$  pellet resting on a ceramic cylinder, which is inserted into a copper tube in a sealed Swagelok assembly. Insulation is placed on one side of the fuel pellet to increase thermal conductivity through the copper on one side. It is worth noting that the surface side area introduced to the cold bath (quenching) is 1/6 of the total pellet surface area; a “contact area” will refer to this area throughout the following



**Fig. 3.** (a) Temperature readings of thermocouples from a pellet submersion experiment of a  $\text{UO}_2$  pellet. Note that, the step changes seen in the plot are due to malfunction of thermocouples in the salt bath at the presented temperature. After reaching the appropriate temperature the sample was then quenched into the cold bath, hence the sudden and smooth drop. (b) Hirox image of a successfully cracked  $\text{UO}_2$  pellet.

**Table 1**

The submersion times and temperatures of each experiment.

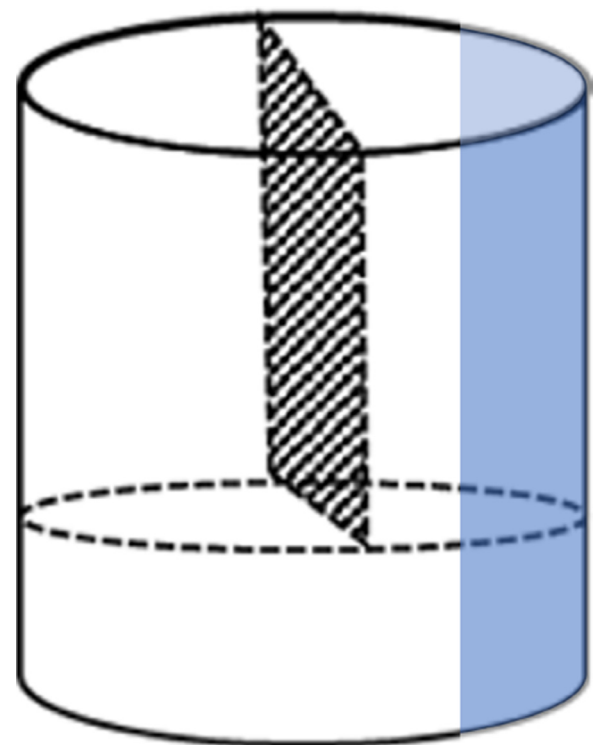
| Pellet ID | Hot bath temp (C) | Density (g/cc) | Capsule hot bath time (min) | Highest pellet temp (C) | Cold bath temp | Capsule cold bath temp | Lowest pellet temperature |
|-----------|-------------------|----------------|-----------------------------|-------------------------|----------------|------------------------|---------------------------|
| U6-71B    | 640               | 8.75           | 6                           | 640                     | *10            | 7                      | 35                        |
| U6-40     | 670               | 9.82           | 7                           | 667                     | -11            | 8                      | 17                        |
| U2-25     | 635               | 9.79           | 8                           | 600                     | -9             | 15                     | 0                         |
| U4-53C    | 700               | 10.11          | 9                           | 662                     | -9             | 15                     | 0                         |
| U4-42     | 565               | 10.31          | 10                          | 585                     | 4              | 10                     | 25                        |
| U5-15A    | 683               | 10.38          | 11                          | 676                     | 5              | 15                     | 10                        |
| U2-49     | 712               | 10.04          | 26                          | 669                     | 5              | 14                     | 10                        |

discussions, see Fig. 4 (for a schematic illustration). The tube assembly is constructed inside of a helium-filled glovebox in order to reduce the oxidation rate of the fuel pellet. A successful experiment is one that manages to induce fracture in the pellet as a result of the sudden temperature decrease from the heating in the hot salt bath to quenching in the cold bath. The thermal shock tests were characterized using two thermocouples attached to the assembly, one attached to the top of the pellet and one attached to the outside of the tubing to measure the temperature difference across the boundary. The inner thermocouple (0.3" diameter) is placed into a small divot drilled into the pellet and sealed with high temperature cement while the outer thermocouple is held in place using a hose clamp. Fig. 3(b) shows an optical microscope image of a successfully cracked sample next to an example of temperature data from the same experiment, Fig. 3(a). Note the rapid decrease in temperature as the sample is quenched in the cold water bath.

In this study, as mentioned, several thermal shock experiments were conducted which were utilized for this work modeling validation, demonstrated in Section 6.

### 3.1. Experimental procedure

The capsule was submerged in the molten salt bath until the pellets temperature reached equilibrium with that of the salt bath, about 11 min on average. A sample experimental temperature profile (which did result in fracture) collected via the LabVIEW software suite is shown in Fig. 3(a). The step shifts noticed in this figure exist due to the malfunction of thermocouples in the salt bath at the given temperature. Once approaching the proper temperature, the sample quenched into the cold bath, hence the sudden and smooth drop. Upon reaching the desired temperature, the capsule was removed from the hot bath and immediately submerged into the cold bath. The capsule remained there until it cooled down to around room temperature. Table 1 shows the times and



**Fig. 4.** the pellets were cut into three parts which included two axial cuts and one radial cut to observe the fracture occurring in the samples. Note that shaded in blue is the contact area introduced to the cold bath (1/6 of the total pellet surface area). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

**Table 2**

Sintering profiles and associated calculated densities for various experiments.

| Pellet ID | Mass (g) | Density (g/cc) | Sintering profile                                         |
|-----------|----------|----------------|-----------------------------------------------------------|
| U6-71B    | 9.05     | 8.75           | 30 min at 350 °C, 10 min at 1800 °C                       |
| U6-40     | 9.29     | 9.82           | 30 min at 350 °C, 12 h at 1700 °C, 5 h cool down to 25 °C |
| U2-25     | 9.41     | 9.79           | 30 min at 350 °C, 8 h at 1700 °C, 5 h cool down to 25 °C  |
| U4-53C    | 9.08     | 10.11          | 24 h at 1790 °C                                           |
| U4-42     | 9.16     | 10.31          | 325 bakeout, 18 h at 1675 °C                              |
| U5-15A    | 9.17     | 10.38          | 325 bakeout, 24 h at 1790 °C                              |
| U2-49     | 9.11     | 10.04          | 325 bakeout, 3 h at 1700 °C                               |

temperatures for each experiment. For select experiments, temperature data for the heating process was inaccurate as depicted with an asterisk in Table 2. However, upon removing the capsule, temperature readings stabilized with minimal variation. This malfunction in thermocouple readings may be caused by an interaction between the copper capsule and the molten salt bath. In the experiments that resulted in fracture which are primarily discussed in the main body of this contribution, physically compromised experiments were omitted from study.

The capsules were left in the fume hood for at least 24 h before they were opened and the pellets were subsequently examined via aided and unaided means. Each pellet was imaged with an optical microscope for micro-level fracturing as well as unassisted user observation for macro-level fracture. The pellets that showed signs of cracking were placed into epoxy resin, sectioned, and polished for further imaging in the scanning electron microscope (SEM). SEM imaged experimental results are shown in Sections A.2.2 and A.2.3 for further study. More details about the experiment are included in the appendix.

#### 4. Multiphysics modeling of thermal shock-induced fracture

This section briefly introduces a multiphysics modeling approach that couple heat transfer, mechanics, and fracture to simulate the fracture behaviour in the  $\text{UO}_2$  pellet under a thermal shock. Similar approaches have been conducted in literature, such as those in Li and Shirvan [7], Bourdin et al. [8]. However, the results of this work model are compared against the aforementioned experimental measurements to demonstrate the efficacy of our modeling approach in predicting the fracture propagation that is consistent with the  $\text{UO}_2$  physical experimentation. The authors utilized the phase-field fracture model [9] that is implemented in the MOOSE framework [10], similar to Wen et al. [11] and Miehe et al. [12,13]. In literature, there are several valuable techniques to numerically capture various fracture behaviors. Some of these successful methods are the discrete element method (DEM), the extended finite element method (X-FEM) [14,15], and the cohesive zone method (CZM) [16,17]. Nevertheless, these methods are usually associated with high computational cost. In X-FEM, the cracks are typically defined as discrete discontinuities that ease the brittle fracture simulations in polycrystalline material [18], while the CZM approach applies a traction-separation method that indicates various fracture behaviors [19,20] and struggles to track the complex fracture evolution that is typically observed in real-world experimental data. Moreover, CZM shows mesh-dependent issues since the crack representations are usually limited to the boundaries of the elements [11].

In comparison, phase-field fracture modeling uses a length scale parameter that is assigned to control the damage band while the discrete fracture surfaces are represented through a diffused damage field. This method has improved simulation capabilities for complex crack patterns and topology, such as branching and coalescence [21–24]. Furthermore, the phase-field model of fracture can explicitly track initiation of the crack interfaces through the

evolution of a phase-field order parameter (i.e., the damage variable) by minimizing the total energy of a system composed of elastic and fracture energies. This approach was first applied in the late 1990s by Francfort [25] and Bourdin [26] and has been recently employed to investigate various fracture behaviors in brittle material [22,24,27] and nuclear materials [28]. Additionally, it has also been implemented to investigate fracture behaviors in  $\text{UO}_2$  [29], which is used in this study.

$\text{UO}_2$  is a fluorite ceramic structure holding a cubic elastic symmetry. Strain energy is decomposed into two separate parts, tensile ( $\psi_{\text{elastic}}^+$ ) and compressive ( $\psi_{\text{elastic}}^-$ ), where only the former contributes to fracture. To implement this methodology, Strain Volumetric and Deviatoric Decomposition were used rather than other decomposition methods proposed, such as those in Clayton and Knap [29], Teichtmeister et al. [30], due to its superior performance for brittle materials. In addition, the strain spectral decomposition method, introduced by Miehe et al. [12,13] works well as demonstrated in Jiang et al. [11]. We encourage readers to consult [11] for more details about the strain spectral decomposition scheme and [10] for a more detailed explanation of the Strain Volumetric Decomposition utilized in this work. Consequently, in this model, the elastic energy will be released in a “fracture energy” manner, i.e.:

$$F_{\text{total}} = \Psi_{\text{elastic}} - \Psi_{\text{external}} + \Psi_{\text{fracture}} \quad (1a)$$

$$= \Psi_{\text{elastic}} - \Psi_{\text{external}} + \mathcal{G}_c \int_{\Omega} \gamma_{\text{crack}} \, d\Omega \quad (1b)$$

where,  $\Omega$  is the system domain,  $\mathcal{G}_c$  the critical energy release rate,  $\gamma_{\text{crack}}$  is a geometrical discontinuous function to represent the crack topology, in which  $\gamma_{\text{crack}} = 1$  represents a fully cracked surface, and  $\gamma_{\text{crack}} = 0$  indicated intact (or “unfractured”) material. In our phase-field fracture model, we utilize the order parameter  $c \in [0, 1]$  to parameterize this function, such that:

$$\gamma_{\text{crack}} \approx \gamma(c; l) = \frac{1}{2l} (c^2 + l^2 |\nabla c|^2) \quad (2a)$$

$$\lim_{l \rightarrow 0} \gamma(c; l) = \gamma_{\text{crack}} \quad (2b)$$

Here  $l$  is the model length-scale that characterizes the width of diffuse interface between cracked and uncracked regions in the model. To smooth this discontinuous function ( $\gamma_{\text{crack}}$ ), the so-called “degradation function” is usually applied to the elastic energy in a way that Eq. (1) would be modified as:

$$F_{\text{total}} = \Psi_{\text{elastic}} - \Psi_{\text{external}} + \mathcal{G}_c \int_{\Omega} \frac{1}{2l} (c^2 + l^2 |\nabla c|^2) \, d\Omega \quad (3a)$$

$$= \int_{\Omega} \psi_{\text{elastic}} \, d\Omega - \int_{\partial\Omega} \psi_{\text{external}} \, d\Gamma + \mathcal{G}_c \int_{\Omega} \frac{1}{2l} (c^2 + l^2 |\nabla c|^2) \, d\Omega \quad (3b)$$

$$= \underbrace{\int_{\Omega} g(c) \psi_{\text{elastic}}^+ \, d\Omega + \int_{\Omega} \psi_{\text{elastic}}^- \, d\Omega}_{\text{degraded elastic energy}} - \underbrace{\int_{\partial\Omega} \tau \cdot u \, d\Gamma}_{\text{external energy}}$$



$$+ \underbrace{\mathcal{G}_c \int_{\Omega} \frac{1}{2l} (c^2 + l^2 |\nabla c|^2) d\Omega}_{\text{fracture energy}} \quad (3c)$$

where,  $\Gamma_{\text{crack}}$  is the crack set/surface.  $g(c)$  is the degradation function that is works to degrade the elastic energy based on the local order parameter. In literature, there are many ways to form this degradation function. Here we utilize a type of degradation function similar to Jiang et al. [11], thus:

$$g(c) = (1 - c)^2(1 - k) + k \quad (4)$$

To avoid numerical issues, we approximate an initial small value of  $k$  ( $\approx 10^{-6}$ ) across the domain, and set  $g(0) = 1$  and  $g(1) = 0$ . The total stress in the damaged material is defined as:

$$\begin{aligned} \sigma &= \frac{\partial F_{\text{total}}}{\partial \epsilon} = ((1 - c)^2(1 - k) + k) \frac{\partial \psi^+}{\partial \epsilon} + \frac{\partial \psi^-}{\partial \epsilon} \\ &= ((1 - c)^2(1 - k) + k) \sigma^+ + \sigma^- \end{aligned} \quad (5)$$

As mentioned above, in this work we are using Strain Volumetric and Deviatoric Decomposition method that describes the orthogonal decomposition of strain tensor in spherical and deviatoric components. Thus, the strain tensor  $\epsilon$  is defined as follows:

$$\epsilon = \epsilon_S + \epsilon_D \quad (6a)$$

$$\epsilon_D = \frac{1}{n} \text{tr}(\epsilon) I \quad (6b)$$

where  $I$  denotes the  $n$ -dimensional identity tensor,  $\epsilon_S$  is the spherical strain tensor, also the strain energy defined as:

$$\psi^+ = \frac{1}{2} \lambda \left\langle \text{tr}(\epsilon)^2 \right\rangle_+ + \mu \epsilon_D \cdot \epsilon_D \quad (7a)$$

$$\psi^- = \frac{1}{2} \lambda \left\langle \text{tr}(\epsilon)^2 \right\rangle_- \quad (7b)$$

where  $\lambda$  is the lame constant and  $\mu$  is the shear modulus. The stress tensors are defined as

$$\sigma^- = \lambda \left\langle \text{tr}(\epsilon) \right\rangle \quad (8a)$$

$$\sigma^+ = \mathbb{C} \epsilon - \sigma^- \quad (8b)$$

where  $\mathbb{C}$  is the fourth-order elasticity tensor. In this work we assume a quasi-static system, therefore the equilibrium equations can be obtained by taking the variations of the total internal energy  $F_{\text{total}}$ , then

$$\nabla \cdot [((1 - c)^2 + k) \sigma^+ + \sigma^-] = 0 \quad (9)$$

where  $\sigma^+$  and  $\sigma^-$  are the Cauchy stress described by Jiang et al. [11]. The damage variable field  $c$  evolves to minimize the total free energy of the system according to the Allen–Cahn equation [31], thus

$$\frac{\partial c}{\partial t} = -L \frac{\partial F}{\partial c} = -L \left( -2(1 - c) H_o^+ + \frac{\mathcal{G}_c}{l} c - \mathcal{G}_c l \nabla^2 c \right) \quad (10)$$

where  $L$  is mobility and  $H_o^+$  is a history variable to prevent crack healing. More detail on the strain-history functional can be found in Miehe et al. [32,33] and Wen et al. [11].

In this work system analysis, the stress develops and evolves due to the thermal shock. Therefore, we couple the phase-field fracture model with the heat conduction equation so that the model can simulate fracture due to thermo-mechanical loading. The temperature variations are the key factors during the thermal shock analysis. The heat conduction is modeled as:

$$k_0 \nabla \cdot \nabla T - \rho C_p \dot{T} + Q = 0 \quad (11)$$

where  $k_0$  denotes the thermal conductivity,  $T$  is the temperature  $C_p$  is the heat capacity,  $\rho$  is density, and  $Q$  is the heat source. The internal stress due to the temperature variations is entirely dependent on the thermal expansion of the system. Thus, the stress tensor ( $\sigma_0$ ) can be defined as

$$\sigma_0 = \mathbb{C}(\epsilon - \alpha \mathbb{I}(T - T_0)) \quad (12)$$

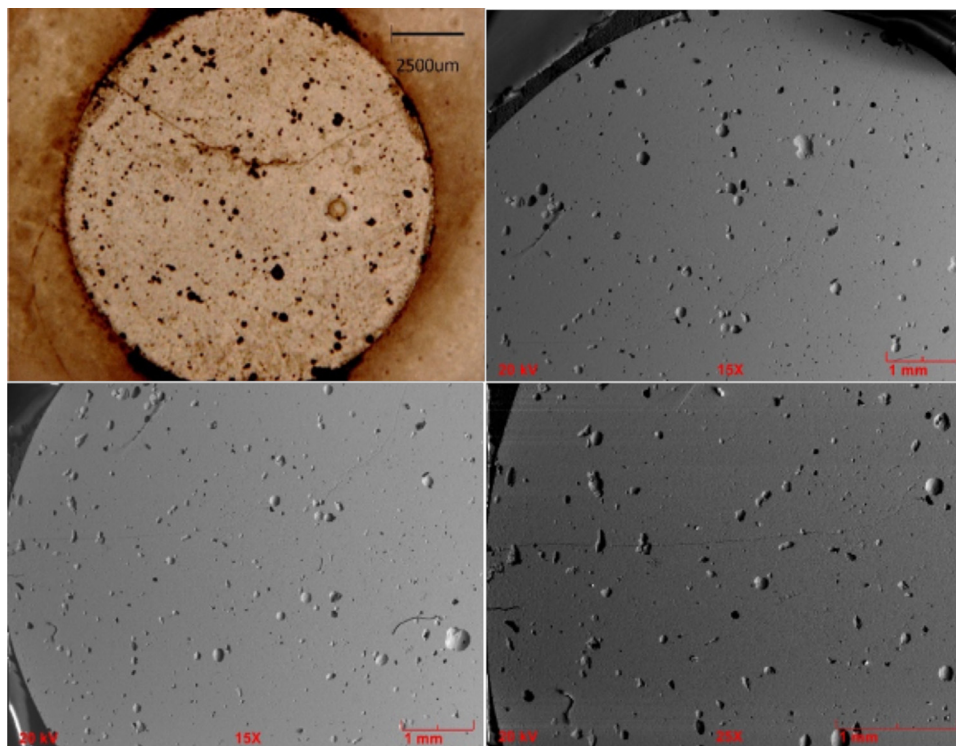
where  $\mathbb{C}$  and  $\epsilon$  are the aforementioned fourth-order elasticity tensor and the strain tensor, respectively. Additionally,  $\alpha$  denotes the thermal expansion coefficient (here, the model take it as a tensor) while  $T_0$  is the reference temperature. Eqs. (9)–(12) are the governing equations for the system; solved simultaneously using the finite element method via the MOOSE framework [34].

## 5. Experimental results

The UO<sub>2</sub> samples were cut axially through the center and then radially through the middle, illustrated in Fig. 4. Numerous optical and SEM images are displayed in Appendix A, representing the final fracture results for each of the pellets, and can be characterized similarly to those shown in Figs. 5 and 6. The pellets were sintered in a mixture of argon and helium gas to prevent oxidation and bakeout was conducted for some pellets to remove any moisture as shown in Appendix A. The pellets were sintered to minimize porosity and the density of the pellets was determined using the Archimedes method also shown in Table 2. The theoretical density of UO<sub>2</sub> is determined to be approximately 10.96 g/cc [35], leading to the density of the pellets being 80–90% of the theoretical density. The variation in density is due to different sintering times, which leads to a change in porosity. A table of densities for various sintering experiments, the sintering processes used, as well as their subsequent quenching thermal quenching results is included in this work in Appendix A.

## 6. Model results and discussion

The phase-field fracture model described in Section 4 was implemented to investigate the fracture behavior of the UO<sub>2</sub> fuel pellet introduced to a thermal shock. The model was utilized to mimic the UO<sub>2</sub> thermal shock experiment as described in Sections 3 and 5. The fundamental model parameters were calculated based on the UO<sub>2</sub> thermomechanical properties, while the other effective parameters were determined based on a parametric study as described below. Due to the instantaneous temperature drop on the pellet surface, a large jump in the stresses throughout the pellet occurred. This behavior is mainly observed on the cooling side face. The rapid temperature change as well as the increased localized stresses are considered the main driving forces for the UO<sub>2</sub> fuel cracking. To obtain initial insights while minding computational costs, we only considered fresh fuel (zero burnup) in this model and then compared these simulated results with the previously described experiment outcomes. The model does not account for creep, i.e. fission gas release effects and grain growth influences were ignored, in the interest of creating a model that is computationally tractable with known and predictable physics. However, the qualitative effects of these phenomena should be considered in parametric study calculations of the energy release rate ( $\mathcal{G}_c$ ) as illustrated throughout this section. The studied domain consists of a 3D fuel pellet with a 10 mm radius, and the axial direction height was set to 10 mm as well. The initial temperature condition of the pellet was set to 750 °C. Only the determined “contact region” of the pellet was introduced to the low-temperature, –10 °C, simulating a thermal shock condition. A Dirichlet boundary condition (BC) for temperature was used to set the thermal bath temperature at the contact area, and Zero flux boundary conditions outside. For the displacement field, a traction-free BC was utilized. While the



**Fig. 5.** The experimental snapshots of various specimens microstructures reveal various porosity (number density and sizes) distributions.

**Table 3**

Parameters utilized in this work for simulation results.

| Property                      | Symbol   | Value              | Unit              | Reference |
|-------------------------------|----------|--------------------|-------------------|-----------|
| Youngs Modulus                | $E$      | 358                | GPa               | [11,36]   |
| Poissons Ratio                | $\nu$    | 0.23               |                   | [36]      |
| Thermal Conductivity          | $k$      | ~5                 | W/m K             | [2]       |
| Heat Capacity                 | $C$      | ~280               | J/(g K)           | [37]      |
| Fuel Density                  | $\rho$   | 10.97              | g/cm <sup>3</sup> |           |
| Energy Release Rate           | $g_c$    | 80                 | MPa-mm            | This work |
| Thermal expansion coefficient | $\alpha$ | $1 \times 10^{-5}$ |                   | This work |
| Length Scale                  | $l$      | $1 \times 10^{-3}$ | mm                | This work |
| Viscosity                     | $\eta$   | $1 \times 10^{-8}$ | s/mm              | [33]      |

BC for the phase-field evolution was a homogeneous zero flux on the entire boundary of the domain and the crack order parameters. In this study, the swelling and the densification consequence were considered negligibly small as described in Li and Shirvan [7]. The model temperature evolution at the pellet center and right-side points against the experimental thermal shock data to verify the model capability to capture the fracture behavior was compared. This verification was presented in Fig. 7, which demonstrates consistency between the computational model and the experimental temperature behavior. Moreover, the model captured the time evolution of the imposed temperature gradient across the pellet correctly, i.e., the outer surface temperature drops immediately to the temperature of the cold bath while the center temperature reaches the same temperature in 300 s. Note that the counter timer in the experimental plot starts after the preheating process at nearly 430 s, as seen in Fig. 3(a). The basic UO<sub>2</sub> thermomechanical properties utilized in this work are listed in Table 3.

It has previously been shown that the strength of brittle materials, such as UO<sub>2</sub>, is strongly dependent on porosity microstructure [7,11]. Moreover, the porosity size and distribution may change with grain size variations. This trend is more pronounced for larger pore sizes [38]. The experimental work presented here obtained various microstructures containing micro and macro pores in dif-

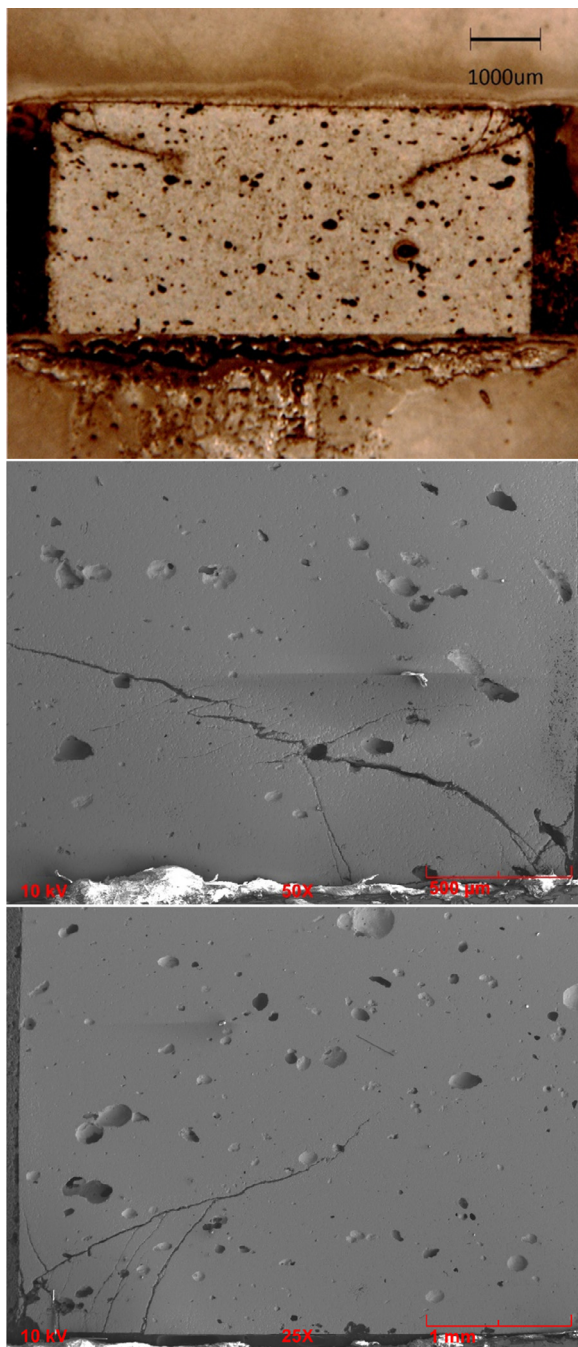
ferent UO<sub>2</sub> specimens, as shown in Figs. 5 and 6, and the Appendix. In some of these specimens, many small spherical pores were almost uniformly distributed in the sample, whereas some exhibited ellipsoidal pores. Since the porosity of the UO<sub>2</sub> specimens undoubtedly affected the fracture mechanics in ways that are not completely understood, various simulations were conducted with different energy release rates ( $\mathcal{G}_c$ ) until a value was settled upon that resulted in a fracture behavior similar to those found in the experiment. It is worth noting that similar methods were utilized experimentally by Doitrand et al. [39] and Henry et al. [40] to determine the strength and fracture toughness of UO<sub>2</sub>. Here, a uniform distribution of porosity is assumed which may not actually be the case, but is still a reasonable assumption. A subsequent method could be developed in future works to account for the size, morphology, and distribution of porosity.

A sensitivity study carried out in this work demonstrates an accordant prediction of the failure behavior and provides consistent information about material strength, fracture toughness, energy release rate, and other model parameters. The outcome of this sensitivity study, as presented in Table 4 and Figs. A.13 and A.14, is used to determine the corresponding model parameters by directly comparing various forms of the predicted failure behavior to the experimental measurements as well as to determine the energy re-

**Table 4**

Detailed description of simulation test case script for the results provided in this section. 45 total simulations were run according to the above schedule.

| Number of test cases | Contact area fraction of the experiment contact area | Energy release rate ( $\mathcal{G}_c$ ) range (MPa-mm) |
|----------------------|------------------------------------------------------|--------------------------------------------------------|
| 9                    | Single Line                                          | 1.5, 5, 10, 30, 50, 80, 100, 150                       |
| 9                    | 1/6                                                  | 1.5, 5, 10, 30, 50, 80, 100, 150                       |
| 9                    | 1/3                                                  | 1.5, 5, 10, 30, 50, 80, 100, 150.                      |
| 9                    | 2/3                                                  | 1.5, 5, 10, 30, 50, 80, 100, 150                       |
| 9                    | 1                                                    | 1.5, 5, 10, 30, 50, 80, 100, 150                       |
| 45                   | Total Number of Test Cases                           |                                                        |
| Selected Test Case   | 100%, same as the experiment test area               | 80 Mpa-mm                                              |

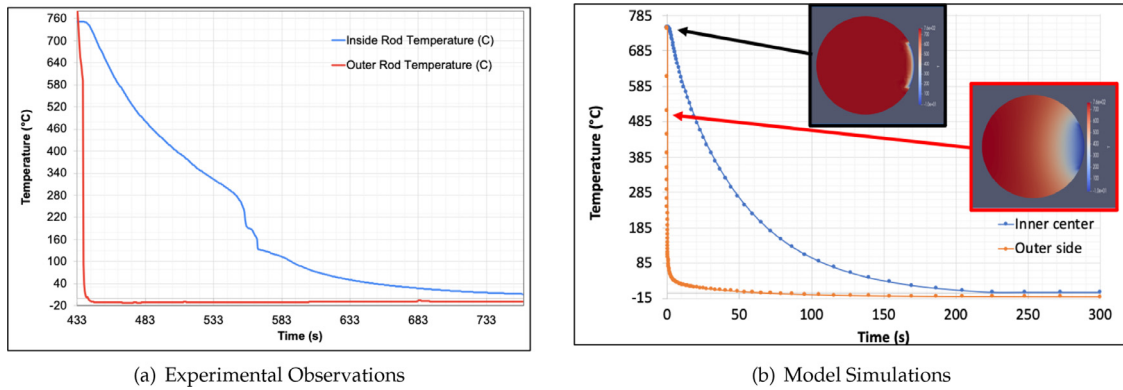


**Fig. 6.** Experimental snapshots of various microstructures reveal various porosity (number density and sizes) distributions as well as subsequent fracture of the pellet. Some porosities are a few micrometers in size or even less, while bigger ones take ellipsoidal shapes. These larger pores have a size of several tens of micrometers.

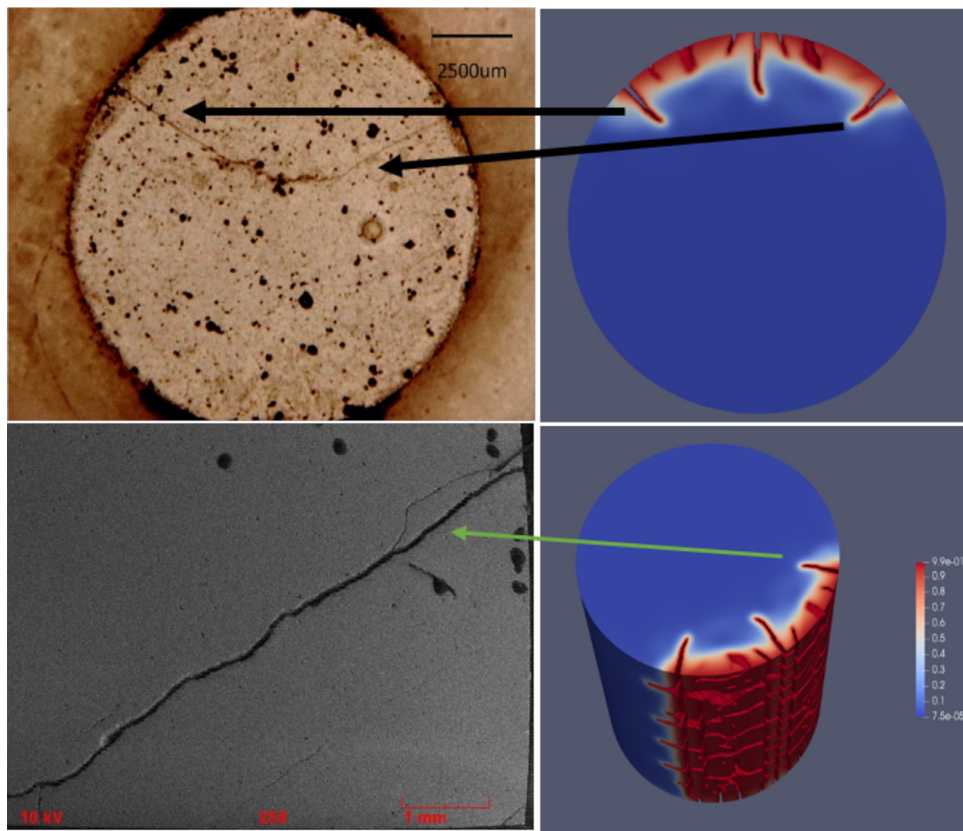
lease rate parameter. To that end, various contact areas configurations and several contact areas of the cooling sides were considered. Some of these contact areas could have varied slightly from those seen in the physical experiments, but these various configurations improve our understanding of the ( $\mathcal{G}_c$ ) parameter influence on the crack formations and evolutions, as well as account for some level of uncertainty in how large the contact region (cooling side area) truly was in the experiment. Additionally, variations on parameters such as ( $\mathcal{G}_c$ ) and the contact region were intended to assist in capturing physical inconsistencies in the experiments, such as insulation, etc that were unaccounted for in the computational model. In all cases, we utilized a range of ( $\mathcal{G}_c$ ) parameters along with the various contact area configurations. The details of the sensitivity study are summarized in Table 4 and some of these simulations are shown with the remainder in Appendix A.

A demonstration of simulations based on what is perceived to be the approximate contact region as the experiment with different  $\mathcal{G}_c$  values varying from 10 to 80 MPa-mm is shown in Fig. A.14. Consistent results between the simulations and the experimental results with  $\mathcal{G}_c = 80$  MPa-mm was observed, as seen in the last row of Fig. A.14 and detailed in Fig. 8. A reasonable agreement is found between the model prediction of the crack pattern and the thermal shock experiment results, as seen in Fig. 8. The model could interpret the formation and evolution of such cracks induced by a thermal shock, particularly the primary radial cracks. There were two major (longer) radial cracks, as illustrated in the hoop and radial stress formation and evolution process in Fig. 9. The cracks formed immediately on the pellet circumferential boundary after the instantaneous drop in the outer temperature (thermal shock). These results demonstrate consistent thermal elastic cracking behavior, wherein the radial cracks are generated by tensile hoop stress. Fig. 9 may be referenced for an improved illustration. It can be observed that the cracks do not propagate inward towards the center of the pellet. Such behavior is anticipated due to the existence of the compression zone that may hinder the cracks propagation, and a similar trend was observed by Li and Shirvan [7]. Compared with the experiment where two major cracks merge in the middle, this discrepancy could be attributed to the porosity of the microstructure causing inconsistent internal compression not anticipated by the computational model, resulting in the crack propagating entirely through the center of the pellet. It is worth noting that some of the tested specimens contain various porosity sizes and distributions and these uncertainties will have an impact on computational fracture predictions. For what is attributed to the same reason, it can be observed that a “middle” crack in the model resulted that did not propagate experimentally. Similarly, it was shown that the crack thickness is consistent with other samples with lower porosity. It is also worth noting that the model correctly reproduces different morphologies of cracks as observed in experiments. One motivation behind future works is to generalize the current model to account for microstructure porosity effects. The elastic strain energy and fracture energy behaviors due to the thermal shock fracture were captured and presented in Fig. 10. As seen from this fig-





**Fig. 7.** The temperature evolution of the pellet outer surface and inner center points. Note that the counter timer in the experiment plot starts after the preheating process at approximately 430 s. The snapshots in the simulation plot show the temperature distribution (colored mapped from  $-10^{\circ}\text{C}$ , dark blue to  $750^{\circ}\text{C}$  dark red) at the initial and mid-time steps. The agreement of the temperature distribution and evolution between the model and the experiment results verified the basic model parameter selection. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

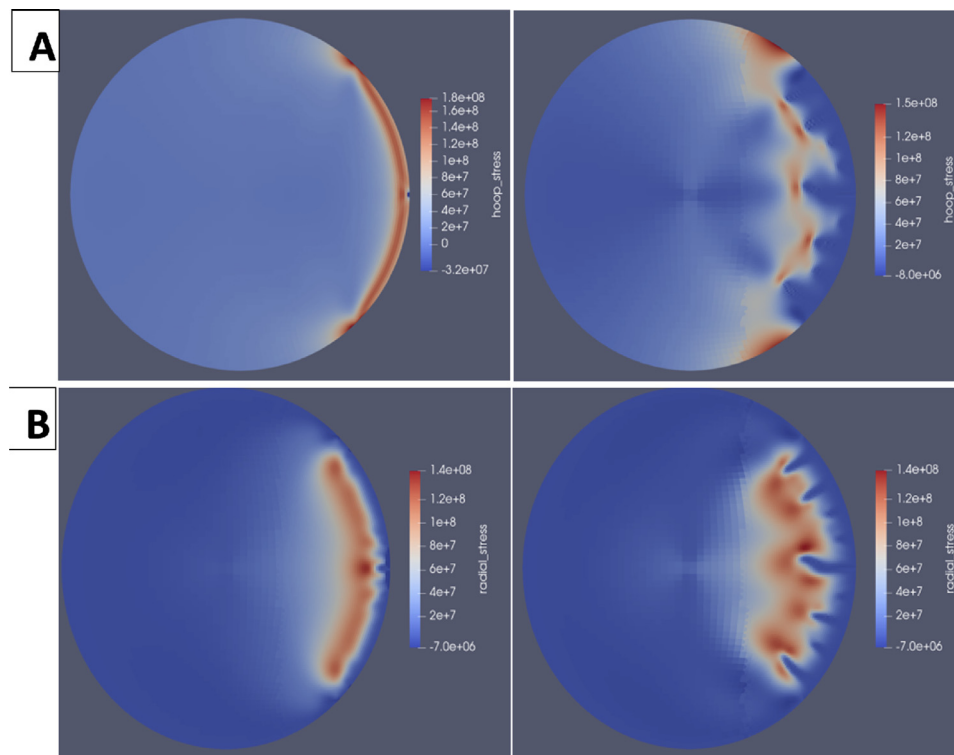


**Fig. 8.** Comparison of final crack patterns induced by thermal shock between this work experiment observations after the thermal shock testing (upper), and (lower) the elected model simulation (colored by the fractured order parameter field distribution). The model could capture the formation and evolution of primary radial cracks. As seen, there are two major (longer) radial cracks that formed immediately on the pellet circumferential boundary after the instantaneous drop in the outer temperature (thermal shock). Moreover, the simulated crack thickness was found to be similar to the experimental observation, as seen in left column. Minor differences between the model and the experiment results were noticed (such as the middle crack shown in the model simulations). These differences can be attributed to the absence of accounting for the effect of underlying microstructure (e.g., size and morphology of pores and grains) in the current model; overcoming such limitations would be the focus of future studies.

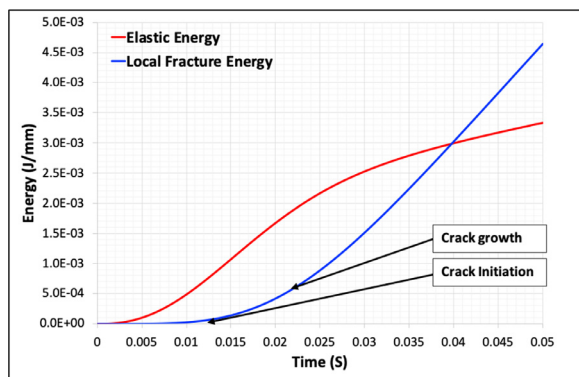
ure, the elastic strain energy rises with increasing the temperature difference between the inner center and outer surfaces (see Fig. 7). As understood from this figure, the cracks immediately appear on the pellet outer surface at about  $10^{-2}$  s right after the instantaneous drop in temperature. After some areas reached a full damage state (fully cracked areas), the rate of increasing of elastic strain declines as it is employed to generate new fracture surfaces, as de-

scribed by Griffith [41]. Furthermore, the figure demonstrates that the local fracture energy increases with crack growth as expected. It is anticipated that this increase in the local fracture energy to be suppressed if the simulations ran long enough to capture the compressive loading in the central region as seen in Li and Shirvan [7].





**Fig. 9.** Snapshots illustrate the stresses formation and evolution before and after the cracks initiation and propagation for (upper) hoop stresses (lower) radial stresses. The left column represents the stresses formation right after the instantaneous drop in temperature, while the right column shows the stresses evolution at the mid of the simulation time.



**Fig. 10.** Evolution of elastic strain energy and the formation and growth of the local fracture energy based on fine mesh and length scale parameter of  $10^{-3}$  mm for the elected simulation case; see the last row in Fig. A.14. The elastic strain energy increases with increasing the temperature difference between the inner center and the outer surface (see Fig. 7). The cracks immediately initiated on the pellet outer surface at about  $10^{-2}$  s, right after the instantaneous drop in temperature. The elastic strain energy rate decreases with generating new fracture faces.

## 7. Conclusions and future work

We introduced a new combined experimental and computational approach to investigate thermal shock-induced fracture in  $\text{UO}_2$  pellets. First, we briefly reviewed the experimental method to promote thermal fracture in the  $\text{UO}_2$ . The details of this method appear in Ortega et al. [6]. Second, the experimental data was utilized to parameterize and validate a multiphysics model of fracture. In contrast to other fracture modeling techniques, our model that employs a phase-field fracture approach can predict the crack nucleation site, growth rate, crack path, size, and morphology

given the values of the fracture properties of the material of interest. The energy release rate,  $\mathcal{G}_c$  (or equivalently fracture toughness) is microstructure- and size-dependent, and its value is unknown for  $\text{UO}_2$  and other nuclear fuels. Its value is difficult to estimate both from experiments and first-principles simulations. Lower-scale, first-principles simulations are limited to small sizes and cannot evaluate the actual value for a realistic microstructure. Even experimental studies cannot extract this number directly. They usually utilize either finite-element simulations or employ a few limiting analytical expressions from the linear elastic fracture mechanics to derive this value from their data [39,40]. Here, a set of  $\mathcal{G}_c$  and contract area were able to capture the overall fracture trend of the corresponding experimental data. In fact, we believe that the phase-field fracture employed here presents a new method for calculating the energy release rate  $\mathcal{G}_c$  /fracture toughness. The model results were obtained by solving the coupled equations using the finite element method implemented in MOOSE. The model predictions demonstrate its capability to simulate the formation and the following growth of the cracks. The model was able to reproduce the experimental temperature profile (see Fig. 7), similar crack morphologies (see Fig. 8), the expected stresses of formation and evolution (see Fig. 9), and the behaviors of elastic and fracture energies predicted by Griffith's theory [41] (see Fig. 10). The importance of the fuel pellets microstructure (e.g., porosity) on the overall kinetics of the fracture formation and evolution were examined qualitatively. A more detailed investigation of the effect of porosity microstructure on macroscopic cracks formation and evolution will be the subject of future studies. Furthermore, the change of elastic and fracture properties with microstructure, temperature, and irradiation conditions will also be accounted for in upcoming investigations. This can be accomplished via coupling our model with the phase-field models of irradiation-induced microstructure evolution [3,42].

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## CRediT authorship contribution statement

**Levi D. McClenny:** Project administration, Software, Writing – original draft. **Moiz I. Butt:** Data curation, Methodology. **M. Gomaa Abdoelatef:** Methodology, Investigation, Validation, Writing – original draft. **Michal J. Pate:** Resources, Formal analysis. **Kay L. Yee:** Formal analysis, Investigation. **R. Harikrishnan:** Data curation, Visualization. **Delia Perez-Nunez:** Resources, Methodology. **W. Jiang:** Conceptualization, Methodology, Writing – review & editing. **Luis H. Ortega:** Resources, Supervision. **Sean M. McDevitt:** Supervision, Funding acquisition. **Karim Ahmed:** Project administration, Methodology, Funding acquisition, Supervision, Writing – review & editing.

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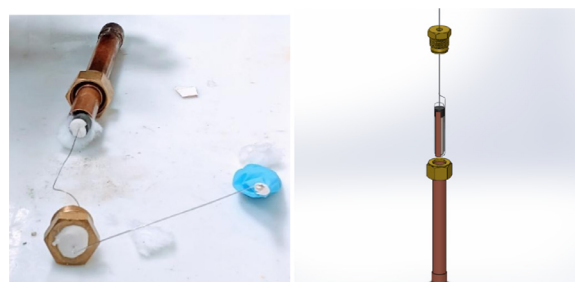
The experimental materials are based upon work supported by the U.S. Department of Energy NEUP IRP program under award agency number DE-NE-0008531. Any opinions, findings, and conclusions or recommendations expressed in this publication are those of the author(s) and do not necessarily reflect the views of the U.S. Department of Energy NEUP IRP program. The modeling and simulation work is supported by the U.S. Department of Energy NEUP program under award agency numbers DE-NE0009134 at Texas A&M University and DE-NE-0008979 at Texas A&M University through a subcontract from the University of Wisconsin-Madison (0000001085). Preliminary simulations of some of the results presented here were obtained as part of a term project for a computational materials science course (MSEN 619/ NUEN 660) offered at Texas A&M University. Portions of this research made use of Idaho National Laboratory computing resources which are supported by the Office of Nuclear Energy of the U.S. Department of Energy and the Nuclear Science User Facilities under Contract no. DE-AC07-05ID14517.

## Appendix A.

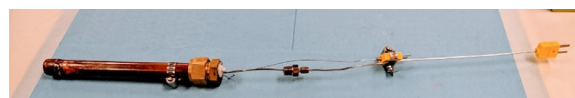
### A1. Experimental setup

A hot molten salt bath and a cold bath connected to a Julabo F-25 chiller were arranged in a fume hood for the experiments in the Fuel Cycle and Materials Lab at Texas A&M University. The molten salt bath consisted of cartescal (ternary carbonate) heat treating salt inside a crucible resting within a heater. The salt composition consists of lithium carbonate, potassium carbonate, and sodium carbonate. The hot bath had to be heated to a temperature above 432 °C to melt the cartescal. In order to reach the 600 °C degrees desired for the experiment, the heater for this bath was turned on three to four hours ahead of time. The cold and the hot bath were turned on at the same time but the cold bath reached the desired temperature of −10 °C to 5 °C much more rapidly. The cold bath contained a 1:1 mixture of ethylene glycol and water connected to the chiller with the same liquid as its working fluid. The liquid in the cold bath was replaced with water for the last three experiments since ethylene glycol can reduce convective heat transfer. Such behavior is undesirable for thermal shocking conditions. Both of the baths had type K thermocouples placed inside of them to record temperatures.

The UO<sub>2</sub> pellets were placed inside a capsule, with a helium environment to prevent oxidation, for each experiment. The capsule



**Fig. A.11.** (left) Insertion of insulation and pellet U6-71B attached to the thermocouple into the capsule. (right) Solidworks rendering of the placement of the pellet in the capsule.



**Fig. A.12.** A finished capsule before it was submerged.

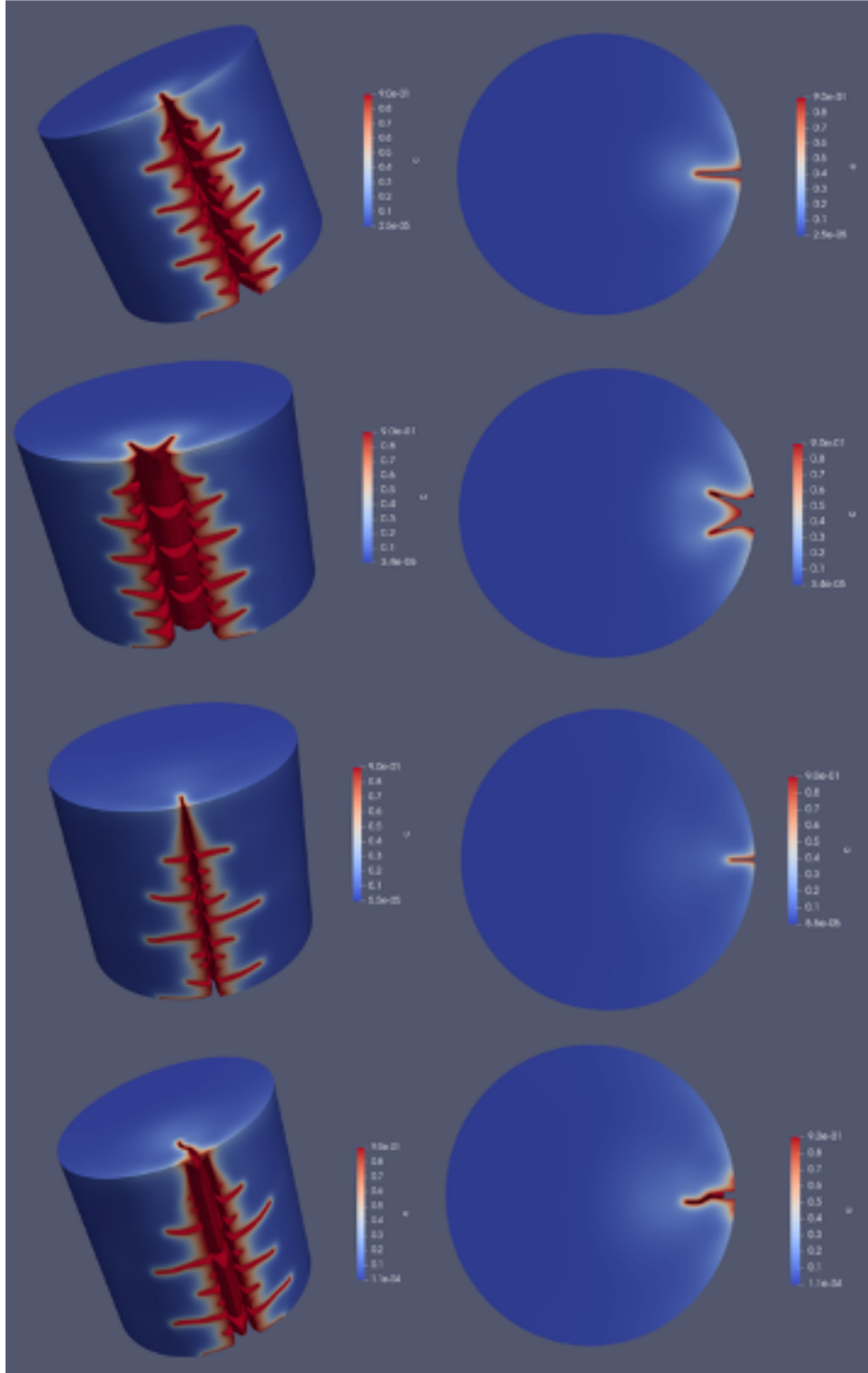
consisted of copper tubing, an end cap, and brass fittings used to seal the top while allowing any sensors to go through, shown in Fig. A.11. For the majority of the experiments, the capsule was a 177.8 mm long copper tube with a diameter of 15.88 mm (5/8in), a copper brazed end cap on the bottom, and a Swagelok brass fitting on the top. The Swagelok fitting had a small hole drilled through the top for the UO<sub>2</sub> thermocouple. Resbond 940HE epoxy held the wires in place and the top was sealed to prevent leakage of coolant mixture into the capsule and interacting with the pellets.

Inside the capsule, a piece of alumina of a diameter similar to the UO<sub>2</sub> pellets (typically 12.7 mm or 1/2 in) was first placed to eliminate axial thermal contact between the bottom of the capsule and the UO<sub>2</sub> so that conductive radial heat transfer was the primary method of heat transfer to occur. This was intentionally designed to form a stress concentration on the contact region to induce fracturing. For most of the experiments, the UO<sub>2</sub> pellet sat on top of the alumina and insulation was placed between the wall of the tube and the pellet. This insulation served to push the pellet to be in contact with one side of the capsule. Fig. A.11(l) shows an example of the insulation as it is being placed inside the capsule. It was placed in such a way that the foil formed thermal contact with the tubing walls on two opposite sides of the pellet. Fig. A.11(r) shows the SOLIDWORKS model of this setup. A type K thermocouple was placed into direct contact with the top center of the pellet and epoxied in place with the Resbond 940HE after being threaded through the hole in the Swagelok fitting. For the last four experiments, a small divot was drilled into the top of the pellet to improve the thermocouples placement on the pellets centerline. A LUNA strain sensor was also attached to the pellet sides using Durabond 952 and threaded through the top for the U6-40 experiment.

After the sensors were attached to the pellet and the Swagelok top fitting, the capsule was assembled and sealed inside a helium filled glovebox to prevent pellet oxidation. Another type K thermocouple was attached to the outside of the capsule using a hose clamp, as shown in Fig. A.12. The two thermocouples attached to the inside and outside of the capsule and the thermocouples in each of the baths were all connected to a NI-cDAQ 9171 measurement chassis recording all temperatures using a custom LabVIEW program on a laptop.

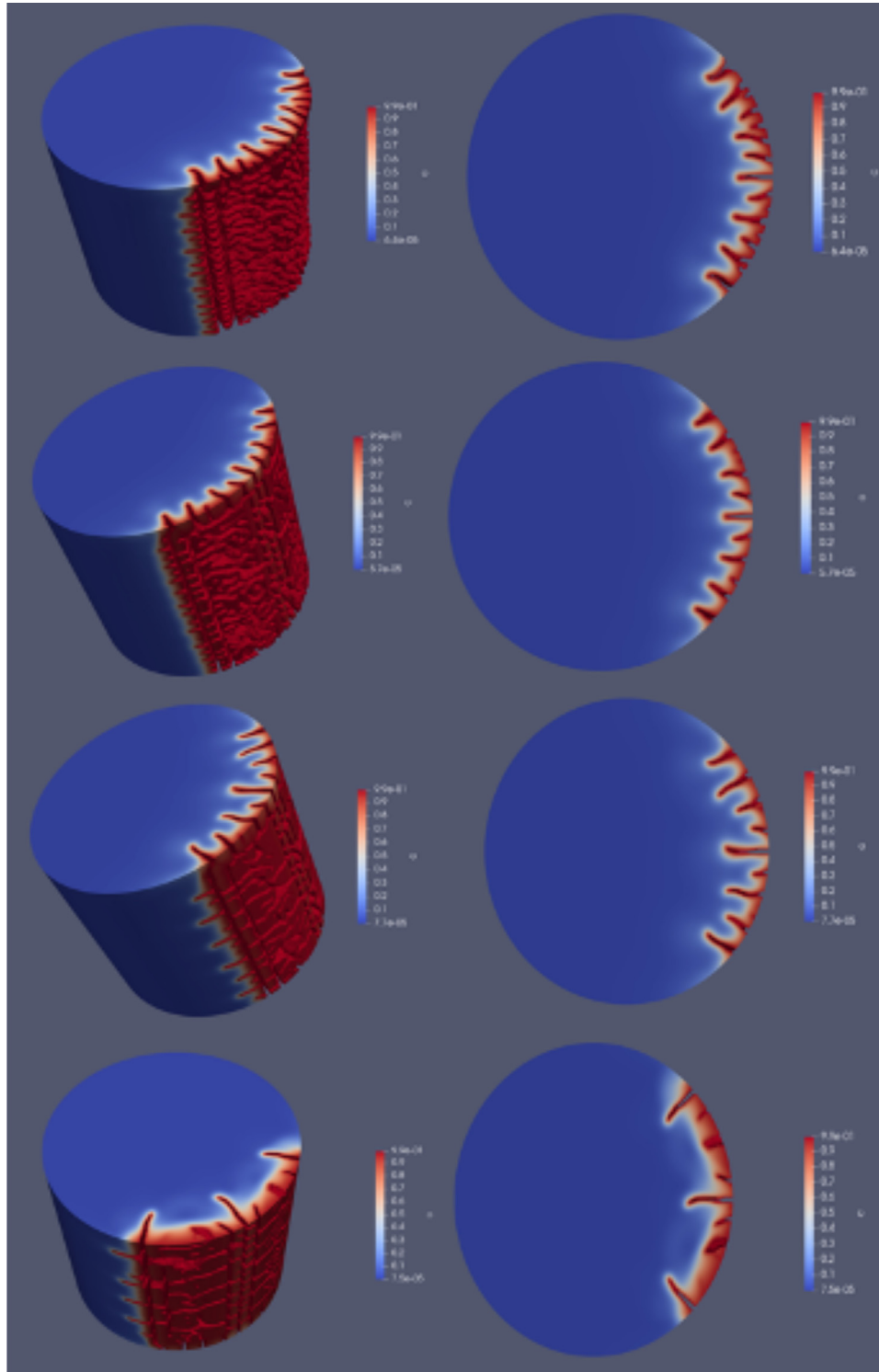
## A2. Additional results

### A2.1. Simulation results



**Fig. A.13.** Snapshots of selected test cases simulations present the effect of different  $g_c$  and different contact areas on the formation and evolutions of the cracks induced by thermal shock in  $\text{UO}_2$  fuel pellets. The first two rows show the cracks behaviors based on  $g_c = 10$  MPa-mm and the last two rows at  $g_c = 30$  MPa-mm. The first and third rows are simply a single line contact area, while the second and fourth ones are established on a wider contact area, almost 1/6 of the actual experiment contact area.

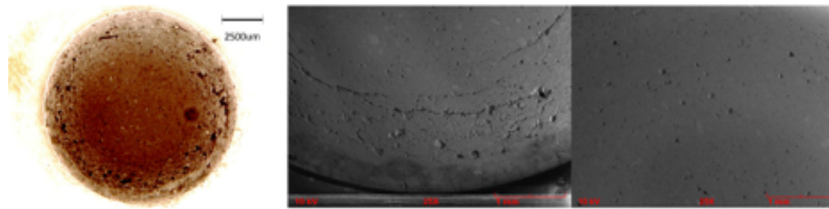




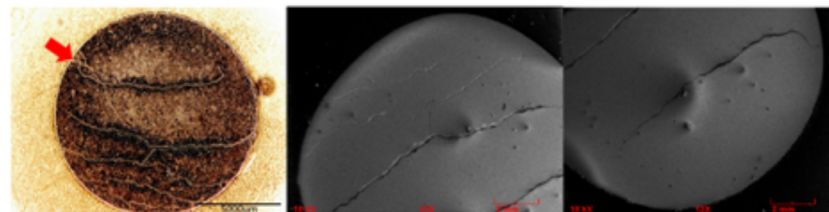
**Fig. A.14.** Snapshots of selected test cases simulations manifest the effect of  $g_c$  values, with the case of actual contact area, on the formation and evolutions of the cracks induced by thermal shock in UO<sub>2</sub> fuel pellet. The damage shape evolutions with  $g_c = 10$  MPa-mm was seen in the first row. While the elected evolution with  $g_c = 80$  MPa-mm, which was found to be well-matched with the experimental results, is presented in the last row. The second and third rows show the simulations with  $g_c = 30$  MPa-mm and  $g_c = 50$  MPa-mm, respectively.

## A2.2. Radial fracture images

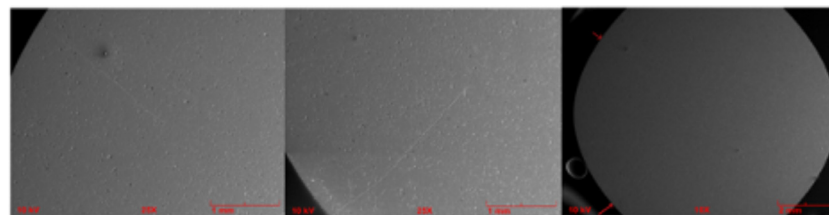
Figs. A.15 –A.22.



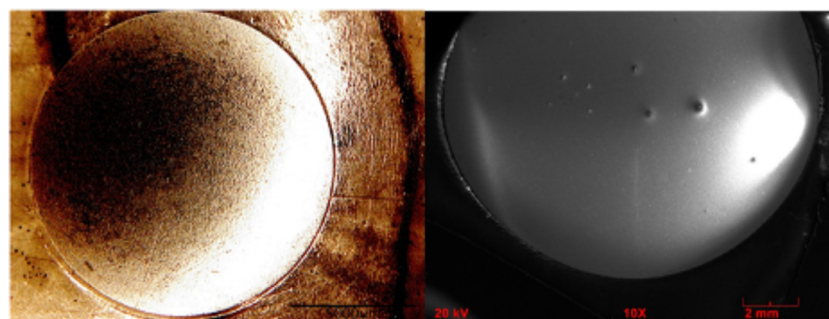
**Fig. A.15.** The experimental snapshots of U66-71B radial cut microstructure reveal various porosity (number density and sizes) distributions. Some porosities have a few micrometers size, or even less, while bigger ones take ellipsoidal shapes. These larger pores have a size of several tens of micrometers. A more detailed investigation of the effect of porosity microstructure on fracture formation and evolution is a part of future studies.



**Fig. A.16.** The experimental snapshots of U5-15A radial cut microstructures.



**Fig. A.17.** The experimental snapshots of U4-53C radial cut microstructures.



**Fig. A.18.** The experimental snapshots of U2-49 radial cut microstructures.

### A2.3. Axial fracture images

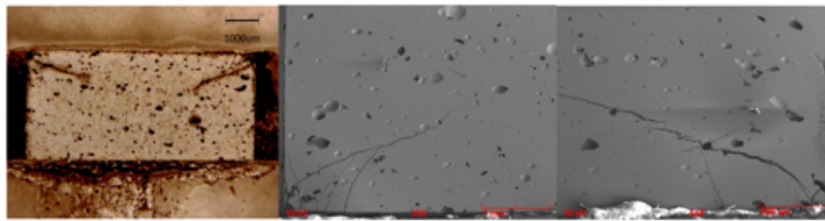


Fig. A.19. The experimental snapshots of U6-40 axial cut microstructures.

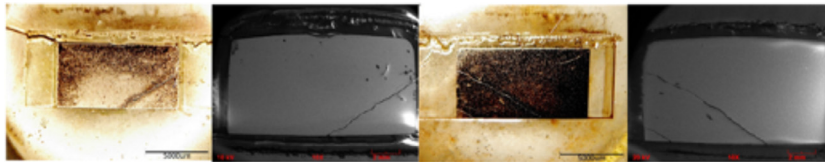


Fig. A.20. The experimental snapshots of U5-15A axial cut microstructures.

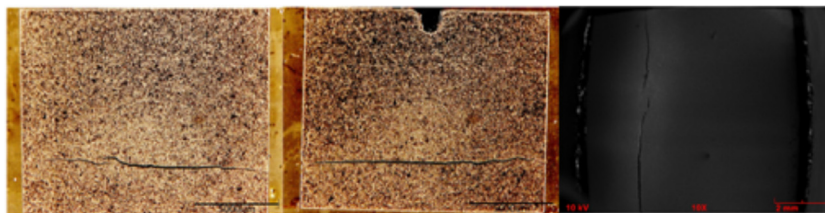


Fig. A.21. The experimental snapshots of U4-53C axial microstructures.

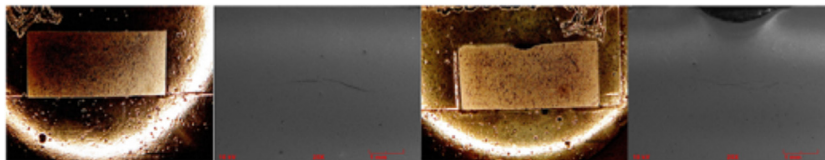


Fig. A.22. The experimental snapshots of U2-49 axial cut microstructures.

### References

- [1] L.H. Ortega, B. Blamer, K.M. Stern, J. Vollmer, S.M. McDevitt, Thermal conductivity of uranium metal and uranium-zirconium alloys fabricated via powder metallurgy, *J. Nucl. Mater.* (2020) 151982.
- [2] F. Badry, R. Brito, M.G. Abdoelatef, S. McDevitt, K. Ahmed, An experimentally validated mesoscale model of thermal conductivity of a  $\text{UO}_2$  and BeO composite nuclear fuel, *J. Mater.* 71 (12) (2019) 4829–4838.
- [3] M.G. Abdoelatef, F. Badry, D. Schwen, C. Permann, Y. Zhang, K. Ahmed, Mesoscale modeling of high burn-up structure formation and evolution in  $\text{UO}_2$ , *J. Mater.* 71 (12) (2019) 4817–4828.
- [4] M.G. Abdoelatef, Mesoscale Modeling of Grain Refinement in Solids, Texas A&M University, 2019 Master's thesis. Available electronically from <https://hdl.handle.net/1969.1/188702>
- [5] B. Spencer, N. Woolstenhulme, L. Emerson, J. Yeh, D. Imholte, C. Hill, D. Chapman, C. Jensen, T. Knight, S. Patnaik, S. McDevitt, L. Ortega, D. Perez-Nunez, H. Ban, Separate-effects validation experiments for models of fracture in ceramic nuclear fuel, INL/CON-19-52564-Revision-0(2019).
- [6] L.H. Ortega, K.L. Yee, D. Perez-Nunez, S.M. McDevitt, C. Steinman, R.R. Schultz, M.L. Dunzik-Gougar, B.W. Spencer, Thermal-shock experiments for separate-effects validation of  $\text{UO}_2$  fuel fracture models, *J. Nucl. Mater.* (2022). To be Published in
- [7] W. Li, K. Shirvan, Multiphysics phase-field modeling of quasi-static cracking in uranium ceramic nuclear fuel, *Ceram. Int.* 47 (1) (2021) 793–810.
- [8] B. Bourdin, J.-J. Marigo, C. Maurini, P. Sicsic, Morphogenesis and propagation of complex cracks induced by thermal shocks, *Phys. Rev. Lett.* 112 (1) (2014) 014301.
- [9] B. Bourdin, G.A. Francfort, J.-J. Marigo, The variational approach to fracture, *J. Elast.* 91 (1) (2008) 5–148.
- [10] D. Gaston, MOOSE: Multiphysics Object-Oriented Simulation Environment, Technical Report, Idaho National Lab.(INL), Idaho Falls, ID (United States), 2014.
- [11] W. Jiang, T. Hu, L.K. Aagesen, Y. Zhang, Three-dimensional phase-field modeling of porosity dependent intergranular fracture in  $\text{UO}_2$ , *Comput. Mater. Sci.* 171 (2020) 109269.
- [12] C. Miehe, F. Welschinger, M. Hofacker, Thermodynamically consistent phase-field models of fracture: variational principles and multi-field fe implementations, *Int. J. Numer. Methods Eng.* 83 (10) (2010) 1273–1311.
- [13] C. Miehe, M. Hofacker, F. Welschinger, A phase field model for rate-independent crack propagation: robust algorithmic implementation based on operator splits, *Comput. Methods Appl. Mech. Eng.* 199 (45–48) (2010) 2765–2778.
- [14] I. Babuška, J.M. Melenk, The partition of unity method, *Int. J. Numer. Methods Eng.* 40 (4) (1997) 727–758.
- [15] N. Moës, J. Dolbow, T. Belytschko, A finite element method for crack growth without remeshing, *Int. J. Numer. Methods Eng.* 46 (1) (1999) 131–150.
- [16] M. Kamaya, Y. Kawamura, T. Kitamura, Three-dimensional local stress analysis on grain boundaries in polycrystalline material, *Int. J. Solids Struct.* 44 (10) (2007) 3267–3277.

- [17] M. Kamaya, M. Itakura, Simulation for intergranular stress corrosion cracking based on a three-dimensional polycrystalline model, *Eng. Fract. Mech.* 76 (3) (2009) 386–401.
- [18] N. Sukumar, D. Srolovitz, T. Baker, J.-H. Prevost, Brittle fracture in polycrystalline microstructures with the extended finite element method, *Int. J. Numer. Methods Eng.* 56 (14) (2003) 2015–2037.
- [19] A. Needleman, Micromechanical modelling of interfacial decohesion, *Ultramicroscopy* 40 (3) (1992) 203–214.
- [20] M. Ortiz, A. Pandolfi, Finite-deformation irreversible cohesive elements for three-dimensional crack-propagation analysis, *Int. J. Numer. Methods Eng.* 44 (9) (1999) 1267–1282.
- [21] A. Karma, D.A. Kessler, H. Levine, Phase-field model of mode III dynamic fracture, *Phys. Rev. Lett.* 87 (4) (2001) 045501.
- [22] A. Karma, A.E. Lobkovsky, Unsteady crack motion and branching in a phase-field model of brittle fracture, *Phys. Rev. Lett.* 92 (24) (2004) 245510.
- [23] H. Henry, H. Levine, Dynamic instabilities of fracture under biaxial strain using a phase field model, *Phys. Rev. Lett.* 93 (10) (2004) 105504.
- [24] M.J. Borden, C.V. Verhoosel, M.A. Scott, T.J. Hughes, C.M. Landis, A phase-field description of dynamic brittle fracture, *Comput. Methods Appl. Mech. Eng.* 217 (2012) 77–95.
- [25] G.A. Francfort, J.-J. Marigo, Revisiting brittle fracture as an energy minimization problem, *J. Mech. Phys. Solids* 46 (8) (1998) 1319–1342.
- [26] B. Bourdin, G.A. Francfort, J.-J. Marigo, Numerical experiments in revisited brittle fracture, *J. Mech. Phys. Solids* 48 (4) (2000) 797–826.
- [27] S. Zhang, W. Jiang, M.R. Tonks, A new phase field fracture model for brittle materials that accounts for elastic anisotropy, *Comput. Methods Appl. Mech. Eng.* 358 (2020) 112643.
- [28] P. Chakraborty, P. Sabharwal, M.C. Carroll, A phase-field approach to model multi-axial and microstructure dependent fracture in nuclear grade graphite, *J. Nucl. Mater.* 475 (2016) 200–208.
- [29] J. Clayton, J. Knap, Phase field modeling of directional fracture in anisotropic polycrystals, *Comput. Mater. Sci* 98 (2015) 158–169.
- [30] S. Teichtmeister, D. Kienle, F. Aldakheel, M.-A. Keip, Phase field modeling of fracture in anisotropic brittle solids, *Int. J. Non Linear Mech.* 97 (2017) 1–21.
- [31] S.M. Allen, J.W. Cahn, Ground state structures in ordered binary alloys with second neighbor interactions, *Acta Metall.* 20 (3) (1972) 423–433.
- [32] C. Miehe, M. Hofacker, F. Welschinger, A phase field model for rate-independent crack propagation: robust algorithmic implementation based on operator splits, *Comput. Methods Appl. Mech. Eng.* 199 (45–48) (2010) 2765–2778.
- [33] C. Miehe, F. Welschinger, M. Hofacker, Thermodynamically consistent phase-field models of fracture: variational principles and multi-field fe implementations, *Int. J. Numer. Methods Eng.* 83 (10) (2010) 1273–1311.
- [34] C.J. Permann, D.R. Gaston, D. Andrš, R.W. Carlsen, F. Kong, A.D. Lindsay, J.M. Miller, J.W. Peterson, A.E. Slaughter, R.H. Stogner, R.C. Martineau, MOOSE: enabling massively parallel multiphysics simulation, *SoftwareX* 11 (2020) 100430, doi:10.1016/j.softx.2020.100430. <http://www.sciencedirect.com/science/article/pii/S2352711019302973>.
- [35] A. Fedotov, E. Mikhnev, A. Lysikov, V. Novikov, Theoretical and experimental density of (U, Gd) O<sub>2</sub> and (U, Er) O<sub>2</sub>, *At. Energy* 113 (6) (2013) 429–434.
- [36] K. Govers, S. Lemehov, M. Hou, M. Verwerft, Comparison of interatomic potentials for UO<sub>2</sub>. Part I: static calculations, *J. Nucl. Mater.* 366 (1–2) (2007) 161–177.
- [37] R. Kavazauri, S.A. Pokrovskiy, V.G. Baranov, A.V. Tenishev, Thermal properties of nonstoichiometry uranium dioxide, *IOP Conf. Ser.* 130 (2016) 012025.
- [38] M. Oguma, Microstructure effects on fracture strength of UO<sub>2</sub> fuel pellets, *J. Nucl. Sci. Technol.* 19 (12) (1982) 1005–1014.
- [39] A. Doitrand, R. Henry, I. Zacharie-Aubrun, J.-M. Gatt, S. Meille, UO<sub>2</sub> micron scale specimen fracture: parameter identification and influence of porosities, *Theor. Appl. Fract. Mech.* 108 (2020) 102665.
- [40] R. Henry, I. Zacharie-Aubrun, T. Blay, S. Chahal, J.-M. Gatt, C. Langlois, S. Meille, Fracture properties of an irradiated PWR UO<sub>2</sub> fuel evaluated by micro-cantilever bending tests, *J. Nucl. Mater.* 538 (2020) 152209.
- [41] A.A. Griffith, VI. The phenomena of rupture and flow in solids, *Philos. Trans. R. Soc. Lond. Ser. A* 221 (582–593) (1921) 163–198.
- [42] K. Ahmed, A. El-Azab, Phase-field modeling of microstructure evolution in nuclear materials, *Handb. Mater. Model.* (2020) 2313–2334.